

1. Project Objective

This project simulates an enterprise infrastructure-compliance data pipeline.

The goal is to ingest raw endpoint and cloud security/compliance information, clean and filter it, perform policy gap analysis, create actionable remediation data, notify downstream teams, report results, and maintain audit evidence.

Learning approach: use AWS services primarily through Python/Boto3 and AWS APIs. Use the console only for basic setup, validation and visual dashboard work.

2. End-to-End Architecture

Raw sources → S3 RAW → Bronze → Silver → Gold → Athena/Glue analysis → compliance gap analysis → remediation queue → SNS/Lambda/SSM → QuickSight reporting. CloudTrail + CloudWatch + KMS provide audit, monitoring and security. Airflow on EC2 finally orchestrates the complete workflow.

3. AWS Services Used

Service	Purpose
Amazon S3	Central data lake for raw, bronze, silver, gold, remediation and report data.
IAM + STS	Identity, roles, permissions and credential verification.
KMS	Encryption and key management.
AWS Config	AWS resource configuration/compliance rules and states.
AWS Systems Manager (SSM)	Inventory, patch/compliance information, Run Command and remediation.
Amazon Inspector	Vulnerability findings and severity information.
AWS Glue	Data Catalog and scalable ETL/PySpark.
Amazon Athena	SQL analysis directly over S3.
AWS Lambda	Serverless validation, transformation and controlled automation.
Amazon EventBridge	Scheduled/event-driven triggers.
Amazon SNS	Critical compliance/remediation notifications.
CloudWatch	Logs, metrics and operational monitoring.
CloudTrail	AWS API audit trail.
QuickSight	Compliance dashboards and management reporting.
EC2	Cloud host for the final Airflow deployment.
Apache Airflow	Final daily pipeline orchestration.
Python/Pandas/PySpark	Data cleaning, transformation and ETL.
Git/GitHub	Version control and project documentation.

4. Practice Dataset

The practice dataset contains **1,020 records × 24 columns** in CSV format. It is synthetic and intentionally mixes compliant/non-compliant and active/non-actionable records so that the business filtering logic can be demonstrated.

Metric	Value
Records	1020
Columns	24
Format	CSV
Primary use	Endpoint/infrastructure compliance pipeline practice

5. Dataset Example

The expanded dataset contains more fields; the schema is listed below.

6. Data Schema / Business Meaning

Column	Meaning
asset_id	Compliance/asset field used for cleaning, filtering, gap analysis or reporting.
user_id	Compliance/asset field used for cleaning, filtering, gap analysis or reporting.
hostname	Compliance/asset field used for cleaning, filtering, gap analysis or reporting.
device_type	Compliance/asset field used for cleaning, filtering, gap analysis or reporting.
device_model	Compliance/asset field used for cleaning, filtering, gap analysis or reporting.
manufacturer	Compliance/asset field used for cleaning, filtering, gap analysis or reporting.
os_version	Compliance/asset field used for cleaning, filtering, gap analysis or reporting.
department	Compliance/asset field used for cleaning, filtering, gap analysis or reporting.
location	Compliance/asset field used for cleaning, filtering, gap analysis or reporting.
status	Compliance/asset field used for cleaning, filtering, gap analysis or reporting.
last_check_date	Compliance/asset field used for cleaning, filtering, gap analysis or reporting.
bitlocker_encryption	Compliance/asset field used for cleaning, filtering, gap analysis or reporting.
antivirus_status	Compliance/asset field used for cleaning, filtering, gap analysis or reporting.
proxy_agent	Compliance/asset field used for cleaning, filtering, gap analysis or reporting.
disk_encryption_software	Compliance/asset field used for cleaning, filtering, gap analysis or reporting.
dlp_status	Compliance/asset field used for cleaning, filtering, gap analysis or reporting.
windows_patching	Compliance/asset field used for cleaning, filtering, gap analysis or reporting.
av_thin_client	Compliance/asset field used for cleaning, filtering, gap analysis or reporting.
network_status	Compliance/asset field used for cleaning, filtering, gap analysis or reporting.
overall_compliance	Compliance/asset field used for cleaning, filtering, gap analysis or reporting.
primary_issue	Compliance/asset field used for cleaning, filtering, gap analysis or reporting.
compliance_check_frequency	Compliance/asset field used for cleaning, filtering, gap analysis or reporting.
device_health	Compliance/asset field used for cleaning, filtering, gap analysis or reporting.
is_action_required	Compliance/asset field used for cleaning, filtering, gap analysis or reporting.

7. S3 Medallion Layout

Layer	Example prefix	Purpose
RAW	raw/endpoint_compliance/	Original source data; never modify.
BRONZE	bronze/endpoint_compliance/	Standardise source and add batch metadata.
SILVER	silver/endpoint_compliance/	Deduplicate, validate, clean and enrich.
GOLD	gold/compliance/	Business-ready KPIs and analytical datasets.
REMEDIATION	remediation/	Actionable data for IT support/automation.
REPORTS	reports/	Daily reports and audit outputs.

8. Cleaning & Filtering Logic

Condition	Treatment
Active + non-compliant + action required	Keep as remediation candidate.
Decommissioned	Exclude from active remediation queue but retain history.
Exited	Exclude from active remediation queue but retain history.
Duplicate records	Keep valid/latest record.
Missing critical identifiers	Quarantine/review.

Condition	Treatment
Multiple failures	Categorise and prioritise.
Stale last-check data	Flag for re-validation.

9. Compliance Gap Analysis

The project compares current endpoint/cloud state against expected compliance controls such as disk encryption, antivirus, proxy, DLP, patching, AWS Config rules and Inspector vulnerability findings. The output is a categorised remediation queue containing the asset, issue, severity/health and action requirement.

10. Remediation & Downstream Delivery

Clean actionable records are written to S3 and can be consumed by IT support teams. Critical records can trigger SNS. Where safe and explicitly configured, SSM or Lambda can perform automated remediation. After remediation, the state is checked again so the pipeline can distinguish resolved and still-non-compliant cases.

11. QuickSight Dashboard

Target KPIs: Total Assets, Active Assets, Compliance %, Non-Compliance %, Action Required, Critical Cases and Remediation Backlog. Target charts: non-compliance by issue, department, location, severity/health and daily compliance trend.

12. Airflow + EC2 Final Stage

After the AWS pipeline works independently, Apache Airflow will be deployed on EC2. The DAG will orchestrate extraction, S3 ingestion, Glue/Athena processing, gap analysis, remediation queue generation, notification and reporting. EC2 will use an IAM role instead of hard-coded AWS credentials.

13. Recommended Repository Structure

```
aws-enterprise-compliance-data-pipeline/
├── data/raw/      data/processed/  data/output/
├── scripts/
├── glue/
├── lambda/
├── sql/
├── airflow/dags/
├── aws/
├── dashboards/
├── reports/
├── docs/
├── requirements.txt
├── .gitignore
└── README.md
```

14. Implementation Roadmap

Phase	Build
1	IAM, STS and Boto3 foundation
2	S3 data lake and raw ingestion
3	Glue Catalog + Athena
4	Python/Pandas data quality and filtering
5	AWS Config
6	Systems Manager
7	Amazon Inspector
8	Glue/PySpark ETL

Phase	Build
9	Lambda + EventBridge + SNS
10	CloudTrail + CloudWatch + KMS
11	QuickSight dashboards
12	Airflow on EC2

15. Interview-Ready Summary

I built an AWS-based enterprise compliance data pipeline that ingests endpoint and cloud infrastructure compliance information into S3, processes and filters it using Python, Athena and Glue, evaluates compliance gaps using AWS-native services, creates actionable remediation datasets, supports notifications and controlled remediation, and publishes compliance analytics through QuickSight. I then orchestrated the end-to-end daily workflow using Apache Airflow running on EC2.

Dataset note: the endpoint compliance data used in this learning project is synthetic.

16. Cost & Resource Discipline

The project deliberately uses paid-capable AWS services to learn the real ecosystem. Resources should be created phase-by-phase, tested, observed, documented and cleaned up. EC2 should be stopped when unused; unnecessary always-on networking such as NAT Gateway should be avoided; Glue, Inspector and QuickSight should be used only when needed; and AWS Budgets/billing alerts should be configured.

17. Complete Data Flow

1) Read raw CSV → 2) Upload to S3 → 3) Add ingestion metadata → 4) Catalog with Glue → 5) Athena quality checks → 6) Clean/deduplicate → 7) Filter lifecycle/non-actionable records → 8) Collect Config/SSM/Inspector information → 9) Transform Silver/Gold → 10) Gap analysis → 11) Remediation queue → 12) SNS notification → 13) Optional SSM/Lambda remediation → 14) Re-check → 15) QuickSight reporting → 16) CloudTrail/CloudWatch audit → 17) Airflow orchestration on EC2.